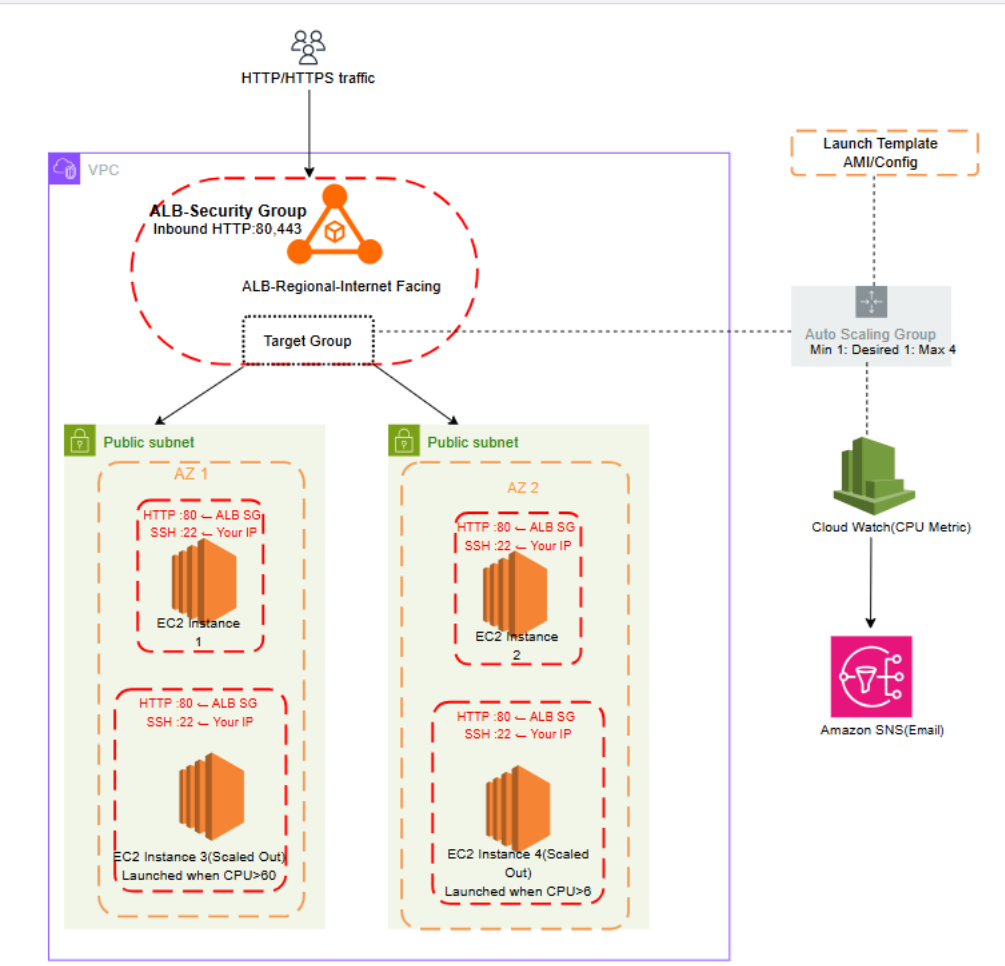
Project: Scalable Web Application using ALB, Auto Scaling, CloudWatch & SNS

This project demonstrates how to deploy a **highly available and scalable web application** on AWS using:

- Amazon EC2 - Hosts the web application
- Launch Template - Defines EC2 configuration
- Application Load Balancer (ALB) - Distributes incoming traffic
- Auto Scaling Group (ASG) - Automatically adjusts number of instances
- Target Group - Registers EC2 instances
- CloudWatch Monitoring - Monitors EC2 metrics
- Amazon SNS - Email Notifications

The system automatically **scales EC2 instances based on traffic load** and **sends email alerts when thresholds are exceeded**.



Step 1: Create a Launch Template

Steps

1. Open **AWS Console**
2. Navigate to **EC2 → Launch Templates**
3. Click **Create Launch Template**

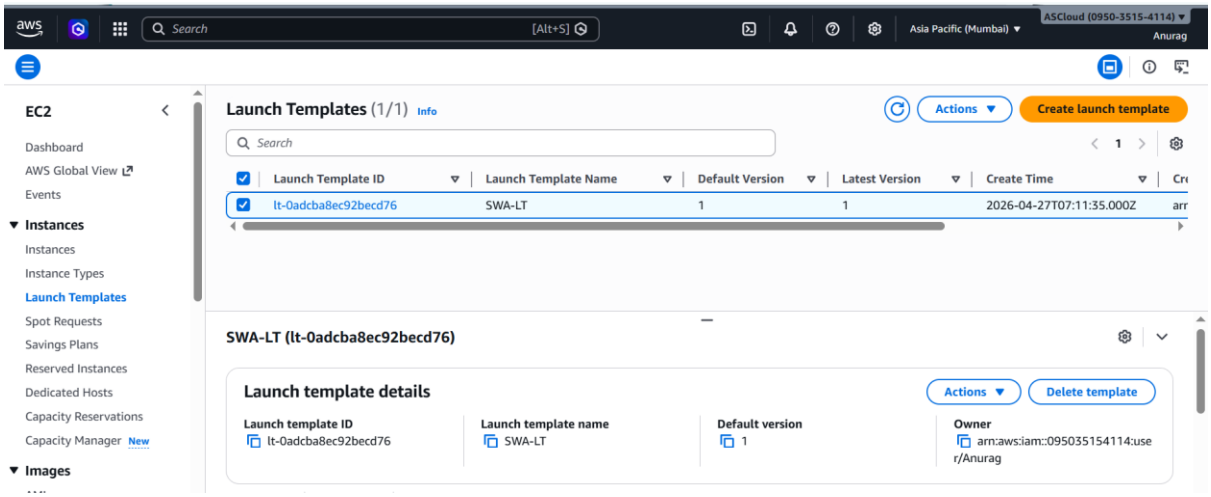
Configure- Name, AMI, Instance Type, Key Pair and SG(Allow 22,80 and 443)

User Data Script

It will automatically installs Apache web server when the instance starts.

```
#!/bin/bash
yum update -y
yum install httpd -y
systemctl start httpd
systemctl enable httpd
echo "<h1>Welcome to Anurag Singh's Scalable Web Application</h1>" >
/var/www/html/index.html
```

Click **Create Launch Template**.



Step 2: Create Target Group

The **Target Group** registers EC2 instances and performs health checks.

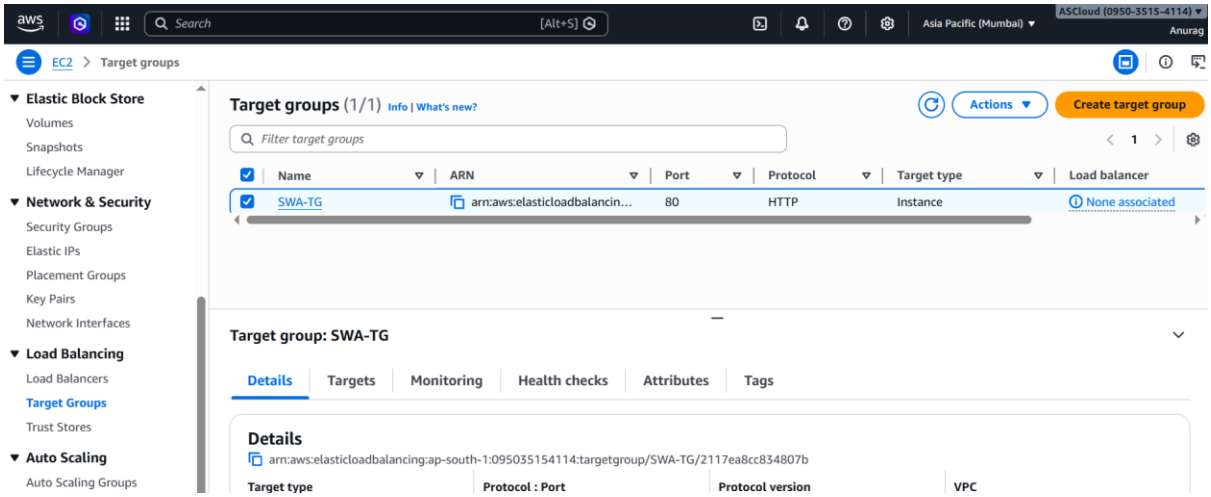
Steps

1. Go to EC2 → Target Groups
2. Click Create Target Group

Configure:

Target Type - Instance
Protocol - HTTP
Port - 80
Health Check Path - /

Click **Create Target Group**.



Step 3: Create Application Load Balancer

The **Application Load Balancer distributes traffic across EC2 instances**.

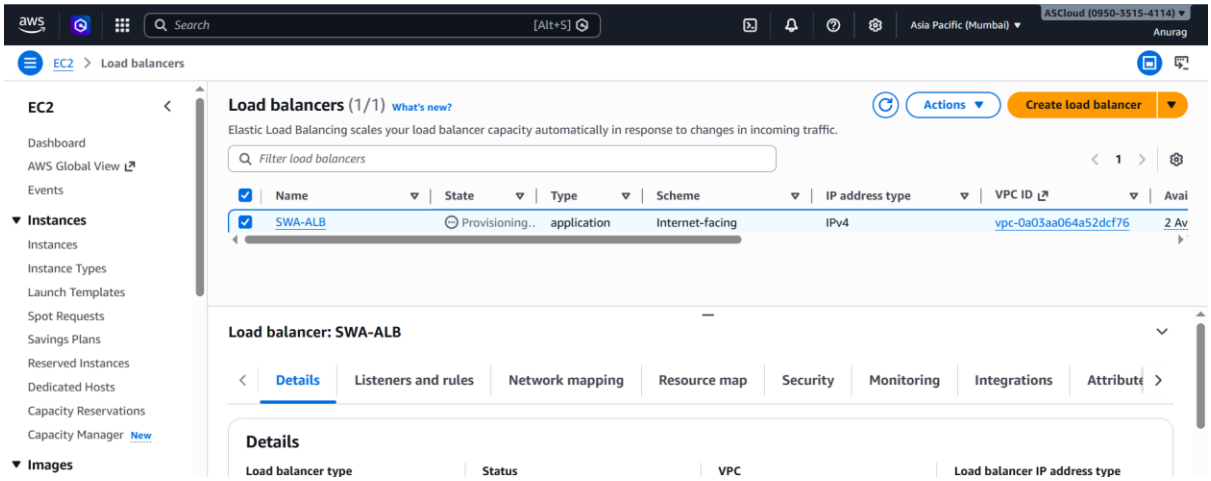
Steps

1. Navigate to EC2 → Load Balancers
2. Click Create Load Balancer
3. Select Application Load Balancer

Configure

Name – SWA-ALB
Scheme - Internet-facing
Availability Zones - Select atleast 2
Listener - HTTP(80)
Security Group - Allow: (Port-80, Source- Internet)
Target Group - Attach the target group created earlier.

Click **Create Load Balancer**.



Step 4: Create Auto Scaling Group

Auto Scaling ensures the application automatically scales based on load.

Steps

1. Go to EC2 → Auto Scaling Groups
2. Click Create Auto Scaling Group
3. Select Launch Template (WebAppTemplate)

Configure Capacity

Minimum Instances - 1
Desired Instances - 2
Maximum Instances - 3

Network Configuration

Select:

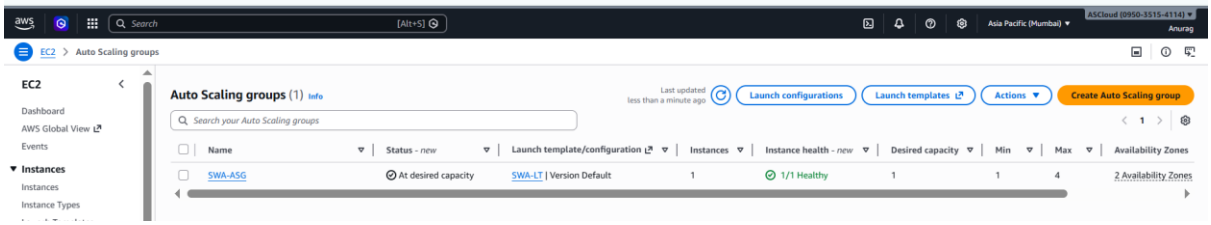
- Your VPC
- Two subnets in different Availability Zones

Attach Load Balancer

Select:

Application Load Balancer

Choose the **Target Group** created earlier.



Step 5: Configure CloudWatch Monitoring and Scaling

CloudWatch monitors EC2 CPU utilization and triggers scaling actions.

Create Scaling Policy

1. Open Auto Scaling Group
2. Go to Automatic Scaling
3. Click Add Scaling Policy

Example configuration:

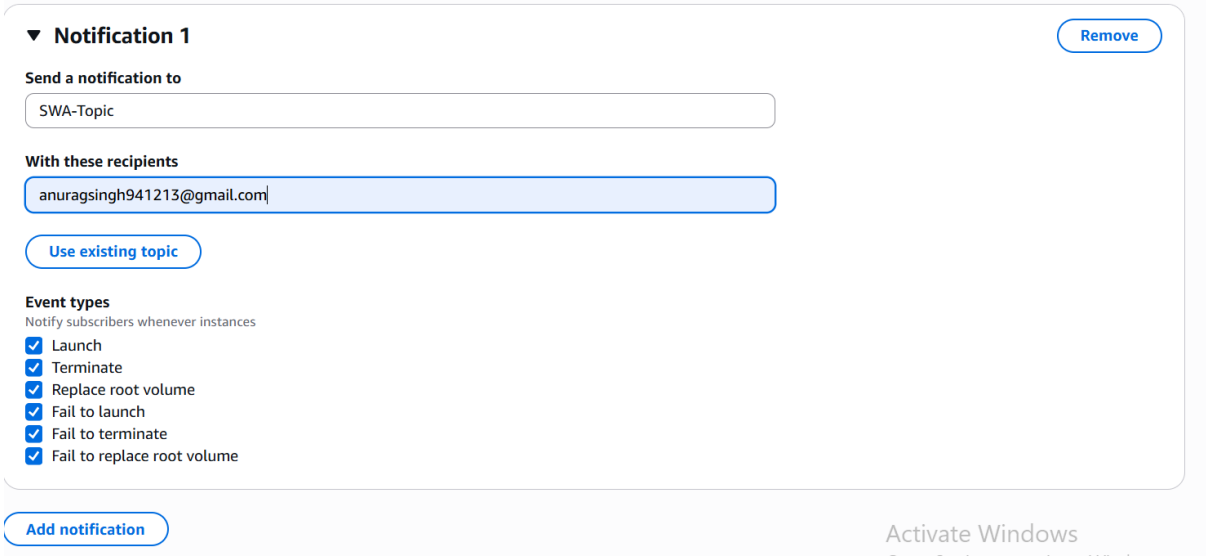
CPU >=50% Add EC2 instance
CPU <50% Remove EC2 instance

CloudWatch automatically monitors:

Step 6: Configure Email Alerts using Amazon SNS & Create SNS Topic

Amazon SNS allows sending email notifications when CloudWatch alarms trigger.

1. Go to AWS Console → SNS
2. Click Create Topic

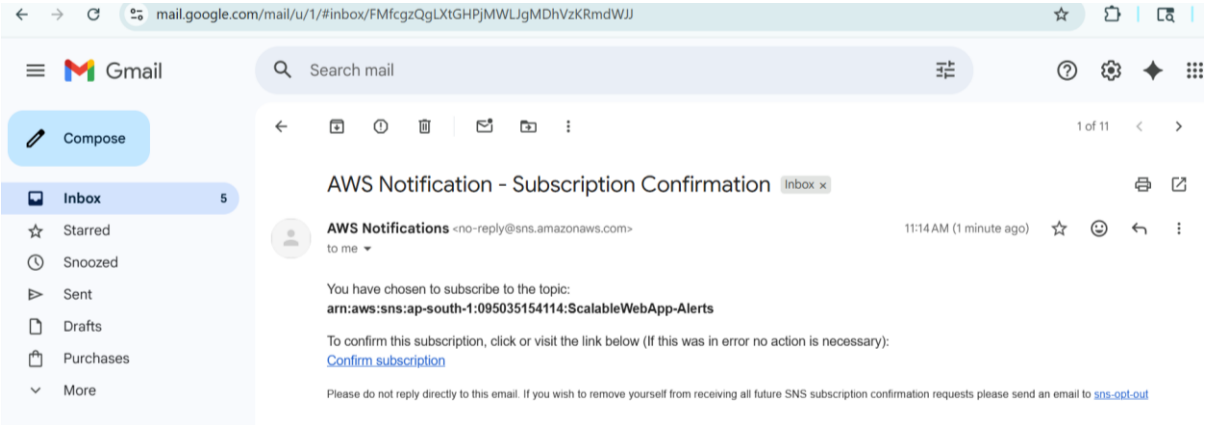


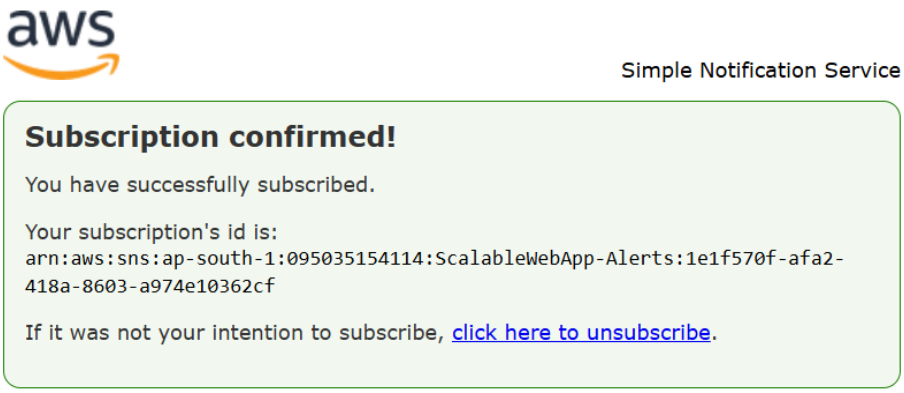
Step 8: Create Email Subscription

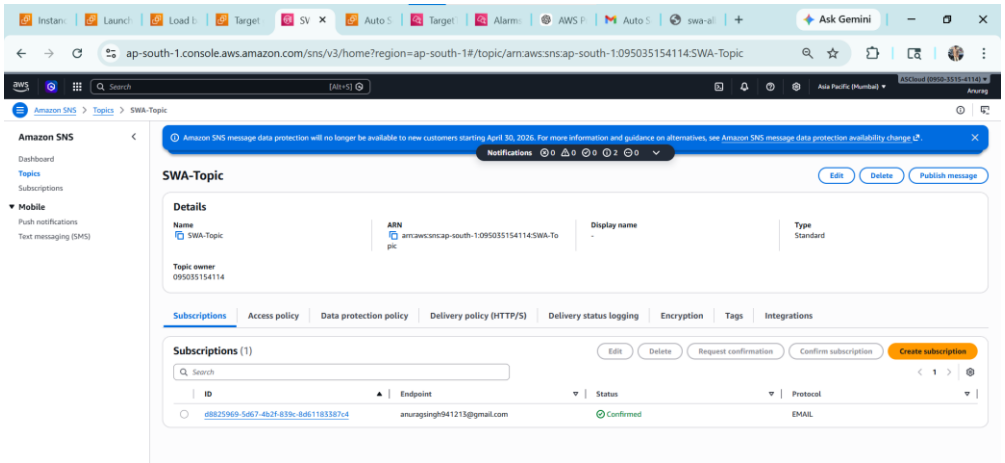
Configure:

Protocol - Email
Endpoint - anuragsingh941213@gmail.com

You will receive an **email confirmation**.
Click **Confirm Subscription**.







Step 9: Connect CloudWatch Alarm with SNS

1. Go to CloudWatch → Alarms
2. Click Create Alarm

Select metric:

EC2 → CPU Utilization

Set threshold: Greater than 50%

Under Alarm Actions select:

Send notification to SNS topic

Choose:

WebApp-Alerts

Step 7: Test the Setup

Access the Application

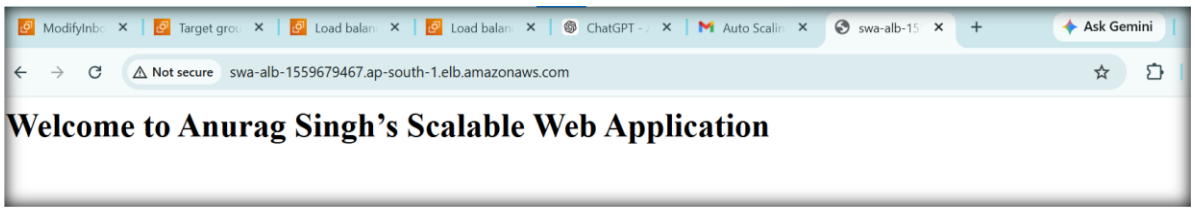
1. Go to EC2 → Load Balancers
2. Copy the ALB DNS Name

Example:

SWA-ALB-1559679467.ap-south-1.elb.amazonaws.com

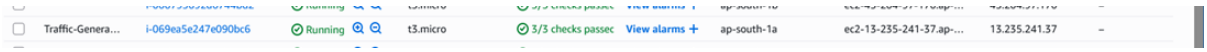
You should see:

Welcome to Anurag Singh's Scalable Web Application



Step 8: Generate Traffic

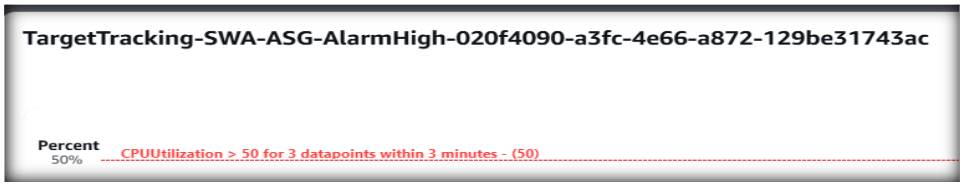
Create another instance and send requests to the DNS of your Application Load Balancer.

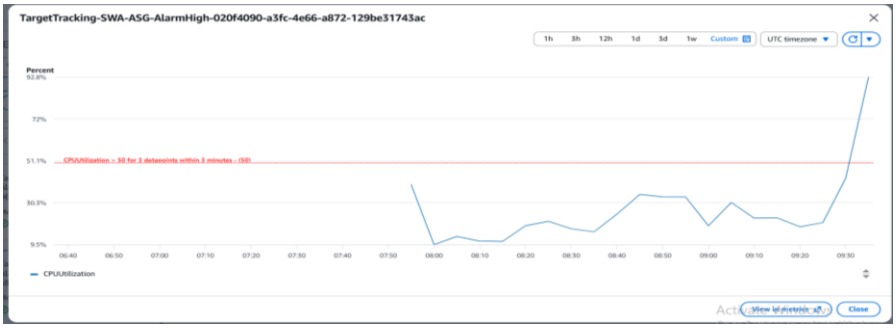


sudo apt install apache2-utils

ab -n 1000 -c 50 http://SWA-ALB-1559679467.ap-south-1.elb.amazonaws.com

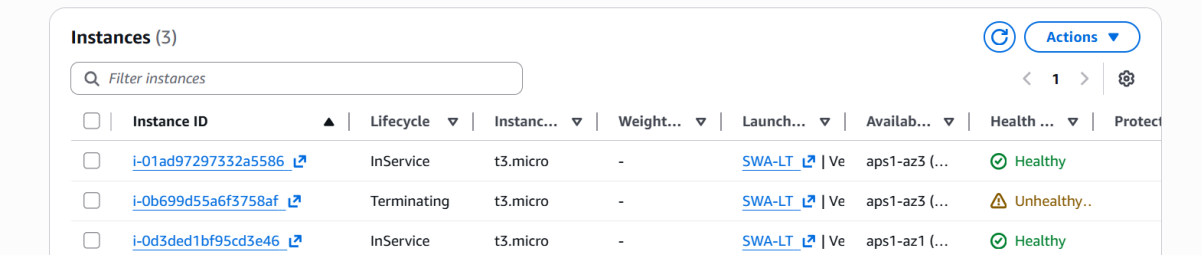
This will increase CPU utilization.





Step 9: Verify Scaling Out

You should see new EC2 instances launching automatically.

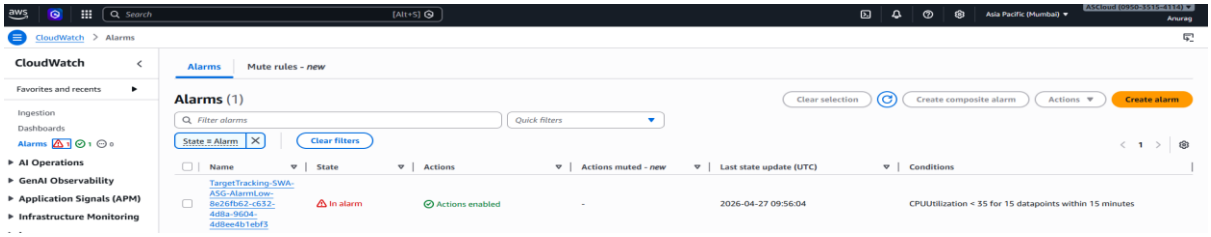


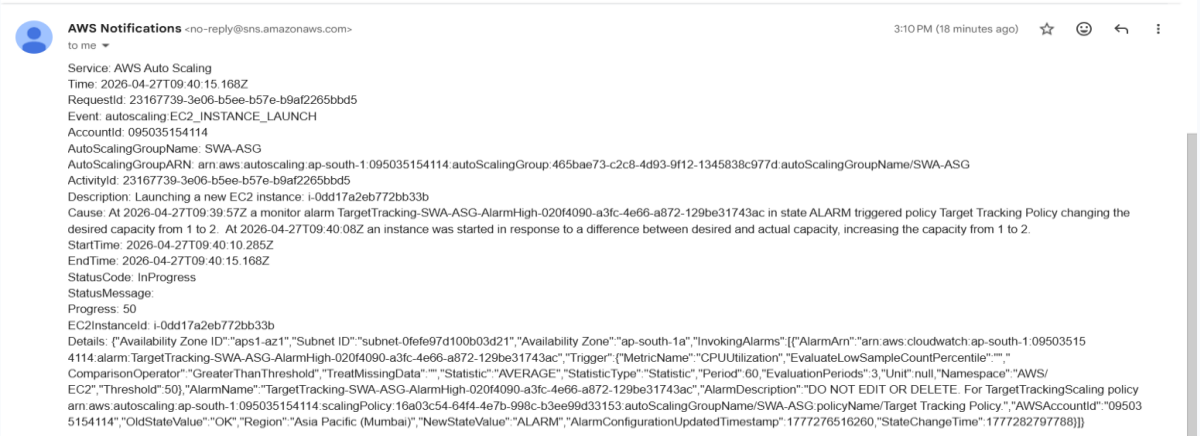
Step 10: Verify Email Alert

When CPU exceeds threshold:

CloudWatch Alarm
↓
SNS Triggered
↓
Email Notification Sent

You will receive an email alert about high CPU usage.

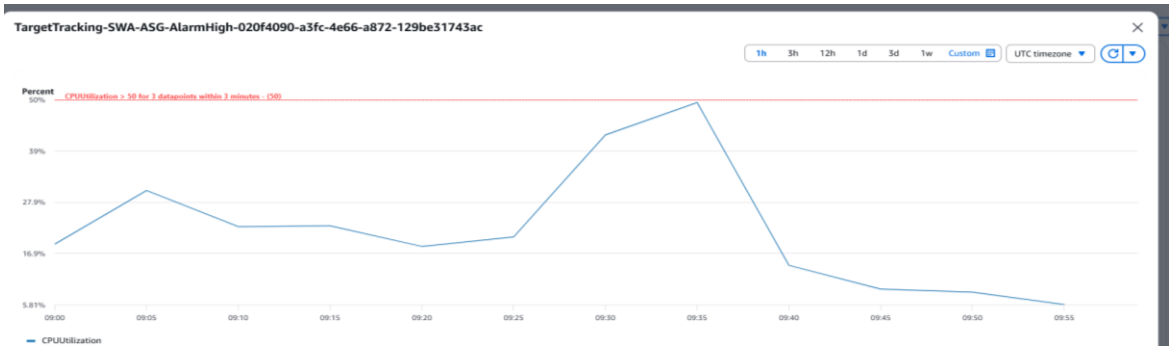


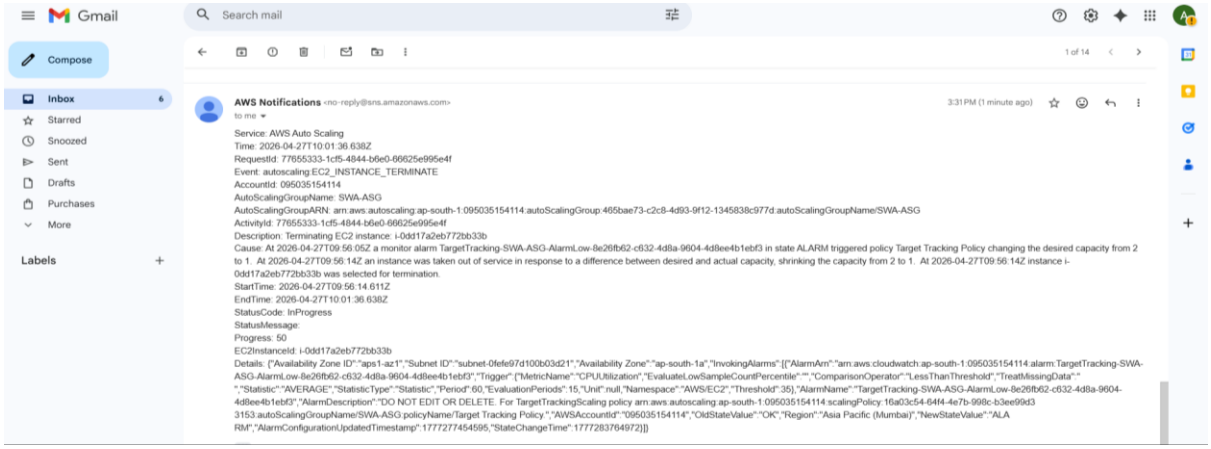


Email about increasing the capacity from 1 to 2

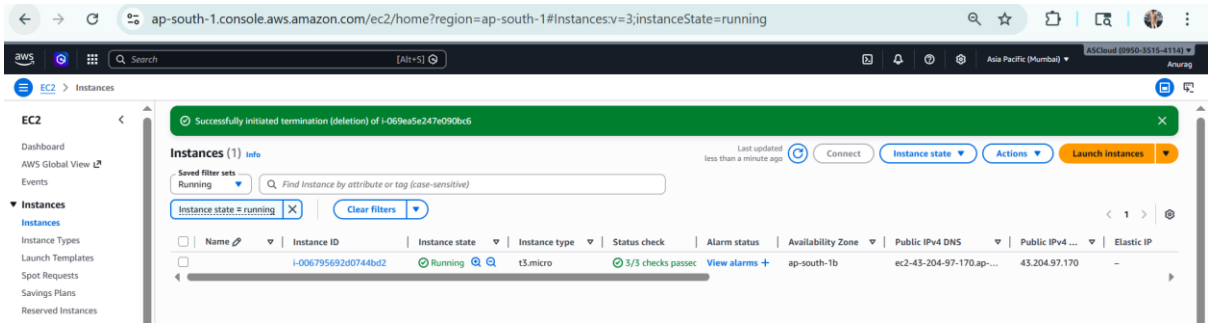
Step 9: Verify Scaling In

Few minutes later the, CPU utilization < 50%





Email described about shrinking the capacity from 2 to 1



Final Architecture Benefits

This architecture provides:

High Availability (Multi-AZ deployment)
Load Balancing (Application Load Balancer)
Automatic Scaling (Auto Scaling Group)
Performance Monitoring (CloudWatch)
Email Alerts (SNS Notifications)